

OLD LAYING HEN: FREEZER TO PLATE

4 to 6 lb bird, 6 qt electric pressure cooker (Instant Pot Pro Plus settings shown). Roseburg OR, ~500 ft: no altitude adjustment. Full research: github.com/jonnystokes/Random-AI-projects

1. TIMELINE (about 4 days, ~30 min hands-on)

When	Do	Why (short)	Skip it and...
Days 0 to 2	Freezer to fridge, wrapped, on a tray. ~1 day per 4 to 5 lb	Slow thaw prevents thaw rigor (violent contraction of pre-rigor frozen meat); surface stays under 40°F	Counter/warm-water thaw = toughest result. From-frozen cooking fast-thaws inside the pot
Day 2 to 3	Rest thawed bird 24 h more in fridge	Post-thaw rigor resolves; enzymes age the meat	Stiffer, chewier
Day 3	Cut into 8 pieces; back + neck to broth pile	2x surface for brine/acid; breast handled separately; fits under fill line	Brine reaches center slower; breast overcooks
Day 3, 12 to 24 h	Brine + acid marinade (box 2), fridge	Seasons to center, holds juice, softens surface collagen	Bland inside, ~2x juice loss, ~3x shear force (spent-hen study)
Day 3 alt (whole bird)	Inject (box 3), rest 2 to 24 h	Only fast way to season a whole bird's core	Outside-only brine needs 48 h
Day 4, last 1 to 2 h, optional	Breast only: ginger or pineapple 1 to 2 h, OR baking soda 15 to 20 min + rinse	Breast has no collagen to soften; enzymes cut fibers, soda holds water	Dry breast (fine if shredding into sauce)
Day 4	Drain, pat dry, optional 3 to 4 min skin-side sear	Roasted flavor in meat surface and broth	No tenderness loss
Day 4	COOK (box 4)	Collagen to gelatin. Irreversible step	Too short = rubbery; quick release = stringy
Day 4	Rest 20 to 30 min IN the liquid	Meat reabsorbs liquid and seasoning while cooling	Less seasoned, bleeds on the board
Day 4	Broth stage (box 5)	Recovers gelatin/flavor from the 60% of the bird that is not meat	Most of the hen's value in the trash

2. BRINE + ACID MARINADE (per lb of meat; pieces; fridge; 12 to 24 h)

Ingredient	Amount	What it brings	Limit
Salt (table)	1 tsp per lb (2% by weight)	Reaches the core (1 to 2 cm in 12 h, full in 24 to 48 h); fibers hold water (loss 12% -> 4 to 7%); carries dissolved flavors in; mild tenderizing	Over ~3% by weight = firm, cured
Vinegar 5%	1 part vinegar : 3 parts water, to cover (~1.2% acid). Lemon: 1 : 5	pH ~4.5: proteins swell and take up liquid; collagen solubility 9% -> 29%; spent hens: shear cut 2/3, cooking loss 50% -> 39%	Over ~1.5% acid or over 24 h = chalky outer mm
Sugar	1/2 tsp per lb	Balances; rides in with salt; helps a sear brown	More tastes like ham brine
Aromatics	Onion, garlic, bay, peppercorn, herbs	Surface + liquid flavor only (1 to 3 mm deep max)	None
Optional, breast only, last 1 to 2 h	1 tbsp grated ginger OR 1/4 cup pineapple juice per lb	Enzymes cut fibers; keep working up to ~150°F during ramp	Overnight or overdose = paste
Optional, breast only	1/2 tsp baking soda per lb, 15 to 20 min, rinse	Raises pH; breast holds water	No collagen effect; soapy if unrinsed. Never soda ash

3. INJECTION (whole bird or big pieces)

Solution	Amount	Where	When
3% salt in stock or water (1 tbsp + 1 tsp salt per 2 cups); dissolved flavors only: soy, Worcestershire, garlic powder; 1 to 2 tbsp melted butter per cup. Strain.	1 oz per lb	Grid every 1 to 1.5 in: thighs, legs, breast. Massage after	2 to 24 h before cooking. Overnight is best

4. THE COOK

	FAST: high pressure (~2 h total)	BEST: low-temp hold (overnight, hands-off)	EASY: slow cooker / slow-cook mode
Mode	Pressure Cook, HIGH (12 psi) or MAX (15 psi)	Sous Vide 165 to 175°F (or Keep Warm Custom). No pressure. Lid closed, valve VENTING	Slow Cook LOW/HIGH (185 to 209°F); glass lid or valve venting
Setup	Pieces + 1.5 to 2 cups liquid (wine, aromatics), halfway up the pieces. Min liquid 1.5 cups	Legs/thighs/wings in a bag with 1 to 2 tbsp wine + aromatics; OR fully submerged in 1% salted stock. Breast in its own bag	0 to 1/2 cup liquid for meat; cover the bird if you want broth
Time	HIGH: 40 to 45 min intact pieces; 75 to 90 min falls apart. MAX: subtract ~20%	Legs/thighs 12 to 24 h at 165 to 175°F (older bird = longer). Breast 2 to 4 h at 150 to 155°F	Pieces 4 to 6 h LOW; whole 10 to 12 h LOW
Release	NATURAL, 20 to 25 min. Never Quick	n/a	n/a
Breast	Pull at 25 to 30 min for sliceable; else shred into sauce	Separate bag, separate temperature	Pull early or shred
Rest	20 to 30 min in the liquid	Chill in bag, or shred into bag juices	In the juices; reduce juices for sauce
Result	Tender braise, driest of the three, excellent broth	Juiciest and most tender an old hen gets	Tender, moderate juiciness
Done test	Leg meat pulls from bone with light pressure. Resists: +15 min. Strings: went too long	Same; hours of overshoot do little harm	Same
Water loss	Zero (sealed)	Lid closed + venting: small; check every 12 to 24 h. OPEN POT LOSES ~1 CUP/HOUR	Negligible with lid on

5. BROTH STAGE (after pulling meat)

In the pot	Liquid	Setting	Time	Notes
Bones, skin, cartilage, back, neck, feet if any; 1 onion, 1 carrot, 2 celery, 2 garlic; 1 tbsp vinegar	Cover, stay under 2/3 line	Pressure Cook HIGH or MAX	90 to 180 min, natural release (two 90 min runs = richest)	Pressure broth is darker and meatier than simmered. Feet make it gel. Vinegar is for flavor; mineral gain is negligible

REFERENCE TABLES

6. PRESSURE = TEMPERATURE = TIME (sea level; old-hen legs to fork-tender; time roughly doubles every 15°F cooler)

Setting	psi	°F	Time to tender	Juiciness	Note
Sous vide / hold	0	143	~3 days	Best	Pink-juicy style; old bird stays slightly chewy
Sous vide / hold	0	165 to 175	12 to 24 h	Best	The recommended 'best track'
Slow cooker LOW/HIGH	0	185 to 209	4 to 6 h pieces; 10 to 12 h whole	Moderate	Both settings end near a simmer; LOW just takes longer to get there
Simmer / boil	0	212	3 to 4 h	Moderate	45 min boiled = rubbery. 4 h boiled = stewing hen
Instant Pot LOW	5.8 to 7.2	229 to 233	75 to 90 min (est.)	Lower	Rarely useful for meat
Instant Pot HIGH	10.2 to 11.6	240 to 243	45 min intact / 90 min shred	Lowest	Default choice
Instant Pot MAX / stovetop	15	250	35 min intact / 70 min shred	Lowest	Saves ~20% time vs HIGH, nothing else

7. LIQUID LEVEL BY GOAL (6 qt pot, one bird)

Method / goal	Liquid	Height on bird	Why
Pressure, whole bird, meat is the goal	1 cup (6 qt) / 1.5 cups (8 qt) + trivet	Bird above water	Steam cooks it; firmer, sliceable; flavor stays in meat
Pressure, pieces, shreddable	1.5 to 2 cups	Halfway up	Braise; gentler on natural release
Pressure, broth stage	Cover bones	Submerged	Extraction needs contact. Never above 2/3 line, never below 1.5 cups
Slow cooker, meat is the goal	0 to 1/2 cup	Bottom 1/2 in or none	Sealed lid: no evaporation; bird gives ~1 cup of its own juice
Slow cooker, soup or broth	Cover	Submerged	You want the flavor in the liquid
Low-temp hold, bag	1 to 2 tbsp in bag; bath full	Bag submerged	Bath is only a heater
Low-temp hold, no bag	Cover completely, 1% salt (2 tsp per qt)	Fully submerged, required	165°F air is a poor heater; exposed meat dries
Stovetop simmer	Cover, lid partly on	Submerged	Loses 20 to 35% in a few hours: top up

Submerging never toughens meat; it moves flavor into the liquid and makes skin soggy. Salt any liquid the meat sits in to ~1%.

8. WHAT MOVES INTO MEAT

Ingredient	Depth	Practical use
Salt (+ sugar, garlic/onion powder, soy, MSG dissolved in it)	To the center in 24 to 48 h, or during any long hold	Equilibrium or dry brine 1.5 to 2%; injection
Acid	A few mm	Surface tenderizing on pieces; mild in the pot (wine, tomato)
Herbs, spices, oil-soluble aromatics	1 to 3 mm even after 24 h (wine: under 1 mm in 18 h)	Put them in the sauce and broth, not the marinade
Cooking liquid, on cooling	Reabsorbed as meat cools	Always rest and shred in the liquid

9. INSTANT POT PRO PLUS QUICK SETTINGS

Item	Value
Pressure levels	LOW 5.8 to 7.2 psi (229 to 233°F) HIGH 10.2 to 11.6 psi (240 to 243°F) MAX 15 psi / 95 to 115 kPa (250°F). MAX only in Pressure Cook and Canning
Manual pressure cook	Pressure Cook > Pressure field (Low/High/Max) > hours/minutes > venting field (NATURAL / Pulse / Quick). Ignore presets
Sous vide	77 to 194°F, 30 min to 99 h 30 min. Lid fully closed, valve VENTING, or lid off. Not a loose lid
Keep Warm custom	77 to 203°F, up to 10 h (a second hold mode for shorter jobs)
Minimum liquid / max fill	1.5 cups / 2/3 line (1/2 for beans or foaming)
Max pressure run	4 h per run; back-to-back runs allowed for broth
Slow Cook	Runs cooler than a Crock-Pot; use HIGH, cut liquid 15 to 20%, at least 3 h
Sealing ring	Replace every 12 to 18 months; keep a spare; one for savory, one for sweet
Time to pressure	8 to 15 min with 2 cups liquid; a full cold pot 20 to 40 min. Ramp and natural release are cooking time too

10. RULES

ALWAYS	Fridge thaw. Salt 12 h minimum. Natural release on meat. Rest in the liquid. Leg-pull test before serving. Water-level check every 12 to 24 h on multi-day holds.
NEVER	Counter or warm-water thaw. Quick release on meat. Pull a pressure cook at 30 min because it 'looks done'. Enzyme marinade overnight. Soda ash. Open pot for a long hold. Fill above 2/3.
IF RUBBERY	It needs MORE time, not less heat. Boiling is not the problem; stopping early is.
IF STRINGY	Went too long under pressure or quick-released. Next time: 10 min less, natural release, or use the low-temp hold.
BREAST	Never juicy on an old hen. Brine or inject; pull early; or plan to shred into sauce. Soda or ginger helps.
SAFETY	Poultry safe at 165°F instantly, or 150°F held 3+ min. Above 140°F is out of the danger zone, so a 165°F hold for 24 h is safe. Delay-start is not for raw poultry.
PRIORITY IF RUSHED	Fridge thaw > 12 h salt > natural release > rest in liquid > acid marinade > cut into pieces > sear > breast treatment.

Sources and full explanations: pressure-cooker-research documents 02, 04, 05, 06 in the repo. Spent-hen vinegar data: Erdem 2025 (PMC12214402). Older-animal collagen: Naqvi et al. 2021, Meat Science.